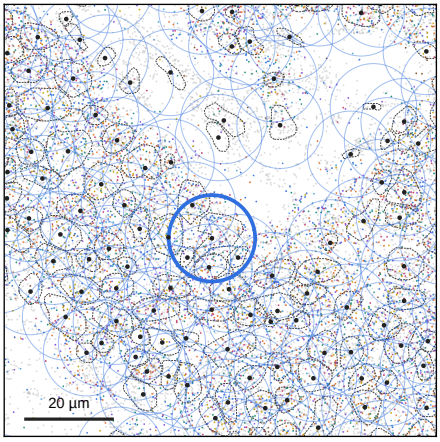
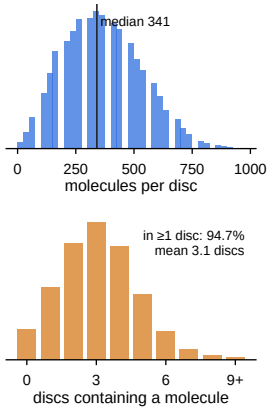
a One 20 µm disc of molecules per nucleus



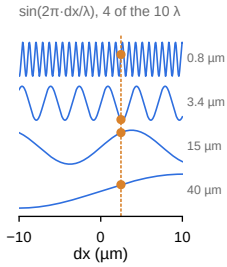
100 µm of the Xenium section; dots: nucleus centroids (from DAPI nuclei); dashed: nuclei; blue: 20 µm discs; colour: gene (top 8)



whole section: 574,852 nuclei; nearest-nucleus spacing median 6.4 µm, so discs overlap

b Each molecule: gene + Fourier-encoded position



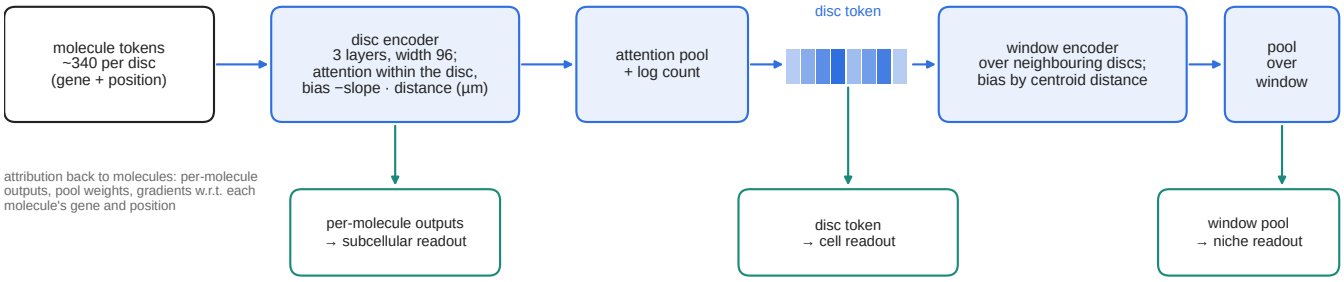


one disc: 545 molecules; offsets from the nucleus centroid; dz from local median z

token = gene_emb(g) + W · [sin, cos](2π · (dx, dy, dz) / λ)

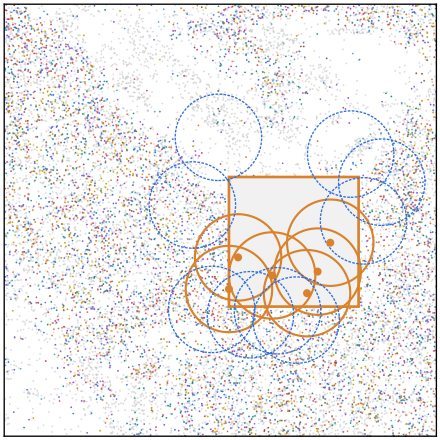
λ = 0.5 … 40 µm, 10 log-spaced (finest ≈ localization precision; coarsest > disc diagonal, so no wrap-around). 3 axes × 10 λ × sin, cos = 60 → W → 96.

c Disc encoder over molecules, window encoder over discs

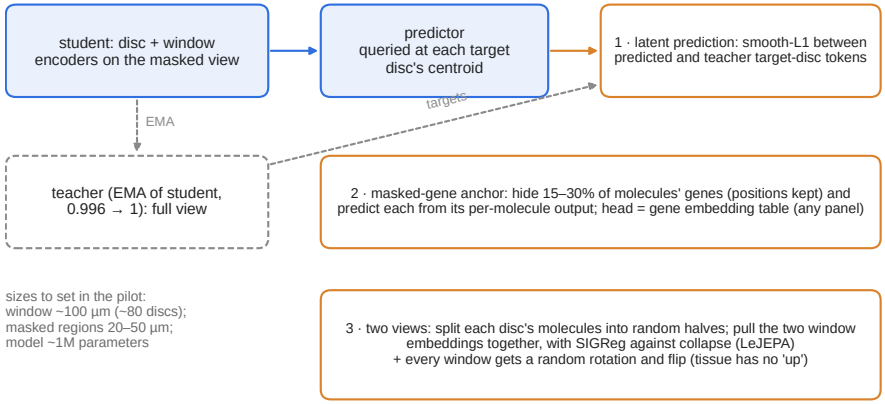


attribution back to molecules: per-molecule outputs, pool weights, gradients w.r.t. each molecule's gene and position

d JEPA training: spatial masking, random rotation, two views



student view: every molecule in the grey region is removed from every disc (dashed blue: neighbours lose part of their disc); orange: discs centred inside are the targets



sizes to set in the pilot:
window ~100 µm (~80 discs);
masked regions 20–50 µm;
model ~1M parameters